

Analyse A Cours 9

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by Lucas Duchet-Annez

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§1 Rappel: Fonctions trigonométriques

§1.1 Equations trigo élémentaires

$$\cos(x) = a, a \in \mathbb{R} \quad (1)$$

1. possède une infinité de solutions $\Leftrightarrow a \in [-1, 1]$
2. si $\cos(x) = \cos(\alpha) = a, \alpha \in \mathbb{R}$
3. alors $x = \begin{cases} \alpha + 2k\pi \\ -\alpha + 2k\pi \end{cases}$

$$\sin(x) = a, a \in \mathbb{R} \quad (2)$$

1. possède une infinité de solutions $\Leftrightarrow a \in [-1, 1]$
2. si $\sin(x) = \sin(\alpha) = a, \alpha \in \mathbb{R}$
3. alors $x = \begin{cases} \alpha + 2k\pi \\ -\alpha + (2k+1)\pi \end{cases}$

§1.2 Exemple d'inéquation

Exemple 1.2.1.

$$\sin(x) \geq \frac{1}{2} \quad (3.1)$$

$$S = \left\{ x \in \mathbb{R} \mid \sin(x) \geq \frac{1}{2} \right\} \quad (3.2)$$

$$x \in S \Leftrightarrow \exists k \in \mathbb{Z} \text{ tq } \frac{\pi}{6} + 2k\pi \leq x \leq \frac{5\pi}{6} + 2k\pi \quad (3.3)$$

$$\Rightarrow S = \bigcup_{k \in \mathbb{Z}} \left(\left[\frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi \right] \right) \quad (3.4)$$

Exemple 1.2.2.

$$|\sin(x)| \leq \frac{1}{2} = \sin\left(\frac{\pi}{6}\right) \quad (4.1)$$

$$S = \bigcup_{k \in \mathbb{Z}} \left(\left[-\frac{\pi}{6} + k\pi, \frac{\pi}{6} + k\pi \right] \right) \quad (4.2)$$

§1.3 Equations en tangente et cotangente

$$\tan(x) = a \quad (5.1)$$

$$\tan(\alpha) = a \quad (5.2)$$

$$x = \alpha + k\pi, k \in \mathbb{Z} \quad (5.3)$$

$$\cotan(x) = a \quad (6.1)$$

$$\cotan(\alpha) = a \quad (6.2)$$

$$x = \alpha + k\pi, k \in \mathbb{Z} \quad (6.3)$$

Exemple 1.3.1.

$$\tan(3x) \leq 1 \quad (7.1)$$

$$\text{Soit } y = 3x, \tan(y) \leq 1 = \tan\left(\frac{\pi}{4}\right) \quad (7.2)$$

$$S_y = \bigcup_{k \in \mathbb{Z}} \left(]-\frac{\pi}{2} + k\pi, \frac{\pi}{4} + k\pi]\right) \quad (8.1)$$

$$S_x = \bigcup_{k \in \mathbb{Z}} \left(]-\frac{\pi}{6} + k\frac{\pi}{3}, \frac{\pi}{12} + k\frac{\pi}{3}]\right) \quad (8.2)$$

$$S_x \cap [0, 2\pi] = \left[0, \frac{\pi}{12}\right] \cup \left[\frac{\pi}{6}, 5\frac{\pi}{12}\right] \quad (8.3)$$

$$\cup \left[\frac{\pi}{2}, 3\frac{\pi}{4}\right] \cup \left[5\frac{\pi}{6}, 13\frac{\pi}{12}\right] \quad (8.4)$$

$$\cup \left[7\frac{\pi}{6}, 17\frac{\pi}{12}\right] \cup \left[3\frac{\pi}{2}, 7\frac{\pi}{4}\right] \quad (8.5)$$

$$\cup \left[11\frac{\pi}{6}, 2\pi\right] \quad (8.6)$$

§2 Formules trigonométriques

§2.1 Formules d'addition

Proposition 2.1.1 (Ptolémée).

$$\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) \quad (9)$$

$$\sin(\alpha + \beta) = \cos(\alpha) \sin(\beta) + \sin(\alpha) \cos(\beta) \quad (10)$$

Preuve. La matrice de rotation d'angle α

$$R_\alpha = \begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix} \quad (11)$$

$$R_\alpha R_\beta = R_\beta R_\alpha = R_{\alpha+\beta} \quad (12)$$

$$\begin{pmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{pmatrix} \begin{pmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{pmatrix} = \begin{pmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{pmatrix} \quad (13.1)$$

$$= \begin{pmatrix} \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) & -\cos(\alpha) \sin(\beta) - \sin(\alpha) \cos(\beta) \\ \cos(\alpha) \sin(\beta) + \sin(\alpha) \cos(\beta) & \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) \end{pmatrix} \quad (13.2)$$

□

Corollaire 2.1.2.

$$\cos(\alpha - \beta) = \cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta) \quad (14)$$

$$\sin(\alpha - \beta) = \sin(\alpha) \cos(\beta) - \cos(\alpha) \sin(\beta) \quad (15)$$

Corollaire 2.1.3.

$$\tan(\alpha + \beta) = \frac{\cos(\alpha) \sin(\beta) + \sin(\alpha) \cos(\beta)}{\cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta)}, \forall k \in \mathbb{Z}, \alpha + \beta = \frac{\pi}{2} + k\pi \quad (16.1)$$

$$= \frac{\cos(\alpha) \cos(\beta) (\tan(\beta) + \tan(\alpha))}{\cos(\alpha) \cos(\beta) (1 - \tan(\alpha) \tan(\beta))} \quad (16.2)$$

$$= \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha) \tan(\beta)} \quad (16.3)$$

$$\cotan(\alpha + \beta) = \frac{\cotan(\alpha) \cotan(\beta) - 1}{\cotan(\alpha) + \cotan(\beta)} \quad (17)$$

Corollaire 2.1.4 (Duplication).

$$\cos(2\alpha) = \cos(\alpha)^2 - \sin(\alpha)^2 \quad (18)$$

$$\sin(2\alpha) = 2 \cos(\alpha) \sin(\alpha) \quad (19)$$

$$\tan(2\alpha) = \frac{2 \tan(\alpha)}{1 - \tan(\alpha)^2} \quad (20)$$

$$\cot(2\alpha) = \frac{\cot(\alpha)^2 - 1}{2 \cot(\alpha)} \quad (21)$$

Corollaire 2.1.5.

$$\sin(\alpha) - \sin(\beta) = 2 \cos\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right) \quad (22)$$

§2.2 Factorisation d'équation trigonométrique simples

Exemple 2.2.1.Résoudre en $x \in \mathbb{R}$

$$\sin(5x) - \sin(x) = \cos(3x) \quad (23.1)$$

$$2 \cos(3x) \sin(2x) = \cos(3x) \quad (23.2)$$

$$\cos(3x)(2 \sin(2x) - 1) = 0 \quad (23.3)$$

$$\cos(3x) = 0 \text{ ou } 2 \sin(2x) - 1 = 0 \quad (23.4)$$

$$x = \begin{cases} \frac{\pi}{6} + k\frac{2\pi}{3} \\ \frac{\pi}{2} + k\frac{2\pi}{3} \end{cases} \text{ ou } x = \begin{cases} \frac{\pi}{12} + k\pi \\ \frac{5\pi}{12} + k\pi \end{cases} \quad (23.5)$$

§2.2.1 Méthode générale

$$a \cos(x) + b \sin(x) = c \quad (24)$$

Exemple 2.2.1.1.

$$\frac{1}{2} \cos(x) + \frac{\sqrt{3}}{2} \sin(x) = \frac{\sqrt{2}}{2} \quad (25)$$

$$\sin\left(\frac{\pi}{6}\right) \cos(x) + \cos\left(\frac{\pi}{6}\right) \sin(x) = \frac{\sqrt{2}}{2} \quad (26)$$

$$\sin\left(\frac{\pi}{6} + x\right) = \sin\left(\frac{\pi}{4}\right) \quad (27)$$

$$\Rightarrow \begin{cases} x + \frac{\pi}{6} = \frac{\pi}{4} + k2\pi \\ x + \frac{\pi}{6} = \frac{3\pi}{4} + k2\pi \end{cases} \quad (28)$$

$$x = \begin{cases} \frac{\pi}{12} + k2\pi \\ 7\frac{\pi}{12} + k2\pi \end{cases} \quad (29)$$

$$\frac{a}{\sqrt{a^2 + b^2}} \cos(x) + \frac{b}{\sqrt{a^2 + b^2}} \sin(x) = \frac{c}{\sqrt{a^2 + b^2}} \quad (30)$$

$$\sin(\alpha) = \frac{a}{\sqrt{a^2 + b^2}}, \cos(\alpha) = \frac{b}{\sqrt{a^2 + b^2}} \quad (31)$$

$$\sin(\alpha + x) = \sin(\beta) = \frac{c}{\sqrt{a^2 + b^2}} \quad (32)$$